Ex no: 2
11/2

Determination of current in circuit using Mesh Analysis

AIM:

To determine the current in circuit using mesh analysis both theoritically and Pratically for a given circuit

APPARATUS Required

S.No	APPARATUS		
1.	Regulated Power supply	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Registors	330Ω, 470Ω, 560Ω	each,
4.	Bread board	-	1
5.			

PROCEDURE:

1. Give connection as per circuit digram
2. Switch on the supply vary the RPS and set a particular voltage
3. Note down the reading of ammeter and note down the readings of all meters.
4. Vary the RPS for different input voltage and note down the readings of all meters
5. Reduce the PPS to its minimum value and switch off the supply
6. Using tabulated values verify kinchoff's laws Pratically and verify theoritically.

Tabular Column

Parameter	Theoretical	Pratical
I_1	1-2 MA	0.5 mA
I_2	4-2 MA	4.8 mA
$I_1 - I_3$	5-4 mA	5.4 mA

Calculations:

$$\begin{bmatrix} R_{11} & R_{12} \\ R_2 & R_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$R_{11} \Rightarrow 330\,\Omega + 470\,\Omega \Rightarrow 800\,\Omega$

$R_{12} \Rightarrow 470\,\Omega \quad R_{21}\,\Omega \quad 470\,\Omega$

$R_{22} \Rightarrow 560 + 470\,\Omega \quad 1030\,\Omega$

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ -5 \end{bmatrix}$$

$\Delta \Rightarrow 800 \times 1030 \; (470)^2$

603100

O/P verified

$\Delta_1 \Rightarrow \begin{vmatrix} 3 & -470 \\ -5 & 3030 \end{vmatrix} \Rightarrow 8090 - 2350 \Rightarrow 740$

$\Delta_2 \Rightarrow \begin{vmatrix} 800 & 3 \\ -470 & 5 \end{vmatrix} \Rightarrow -4000 + 1410 - 2890$

$I_1 \Rightarrow \Delta_1/\Delta \Rightarrow 740/603100 \quad 1.2\,mA$

$I_2 \Rightarrow \Delta_2/\Delta \Rightarrow -2590/603100 \Rightarrow 4.2\,mA$

$I_1 - I_2 \Rightarrow 1.2 - (-4.2) \Rightarrow 5.4\,mA$

S.No	Description	Marks allocated	Marks obtained
1.	Experimental Setup/ connections	20 marks	20
2.	observation/readings	20 Marks	10
3.	calculations/results	20 marks	10
4.	Viva	20 marks	10
5.	Record	20 marks	20
			70

70

Result:

Thus the mesh analysis is verified practically and theoretically. The resultant currents for 3.15 V supply are

a) The current across 330 Ω resistor is 0.5 mA

b) The current across 560 Ω resistor is 4.8 mA

c) The current across 470 Ω resistor is 5.4 mA